

# Low rate speech coding for telecommunications

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*Over the last decade major advances have been made in speech coding technology which is now widely used in international, digital mobile and satellite networks. The most recent techniques permit telephone network quality speech transmission at 8 kbit/s, but there are still demands for even lower rates and more flexible, good quality coding techniques for various network applications. This paper reviews the developments so far, and describes a new class of speech coding methods known as speech interpolation coding which has the potential to provide toll-quality speech coding at or below 4 kbit/s.*

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## 1. Introduction

Low rate speech coding or compression attempts to provide toll-quality speech at a minimum bit rate for digital transmission or storage. The advantages to be gained are bandwidth efficiency, cost reduction, security, robustness to transmission errors, and flexibility in the digital world. The trade-offs [1], dependent on the particular coding technique and the application, are coding delay and distortion, and increased equipment costs.

Throughout the history of digital communications, dating back to the early 1960s and the selection of A-law coding for the first 24-channel pulse code modulation (PCM) systems [2], BT has maintained an active role in speech coding development and standardisation. The early years saw the standardisation of 64 kbit/s PCM, but research into more complex lower bit rate coding schemes was initially inhibited by practical implementation considerations imposed by the semiconductor technology of the day. As a consequence, research into sophisticated low bit rate algorithms did not gather momentum until the late 1970s, and it was not until 1984 that the first world-wide lower bit rate coding standard was achieved. This standard was the International Telegraph and Telephone Consultative Committee (CCITT) G.721 Recommendation for 32 kbit/s adaptive differential PCM (ADPCM), and BT played an important role in the standardisation process.

Since that time, the major advances made in microelectronics and digital signal processor technology have spurred research into increasingly complex speech coding methods. Sophisticated speech coding techniques are now commercially viable, and can be implemented at low cost for more affordable mobile, computer and audio-visual terminals. BT has continued to support codec

standardisation processes via organisations such as the CCITT, the International Telecommunications Union (ITU), the European Telecommunications Standards Institute (ETSI), and the International Maritime Satellite Organisation (Inmarsat), and has achieved notable successes with in-house speech coding technology. Among the latter are the Skyphone™ codec [3] developed at BT Laboratories (BTL), which became the codec chosen by Inmarsat and the Airlines Electronic Engineering Committee (AEEC) as the most suitable codec for aeronautical satellite communications at 9.6 kbit/s [4]. More recently, a low bit rate coder designed at BTL has found application in CallMinder™ [5], a new network-based call answering and message storage service. BTL has also developed the voice activity detector (VAD) [6,7] adopted for use with the global system for mobile (GSM) communications speech coder, and this VAD is attracting interest for use with other mobile radio systems. An efficient VAD is crucial to overall system performance, and reliable operation under high background noise conditions is a key feature of the BTL design.

There was a period in the early 1980s when the need for further speech coding development was brought into question. With the rapid migration to optical fibre transmission that was occurring in world networks, would the potentially vast bandwidth offered by full exploitation of this technology make speech compression unnecessary? This perspective was flawed in that it overlooked the huge growth in demand that would occur for mobile telecommunications. Here, the limited available radio bandwidth makes speech compression essential for the provision of viable services. Furthermore, although the introduction of optical fibre transmission has proceeded apace, bandwidth

constraints remain a significant issue in areas of the fixed network. This is particularly true for international links, whether they are satellite or cable, where speech compression in the form of digital circuit multiplication equipment (DCME) [8] is increasingly employed. Competition from other network operators has also been an important driver for speech coding technology, since more efficient use of the transmission medium permits lower tariffs, leading to increased market share. Another area where speech coding plays a vital role is in the new multimedia services [9]. In these applications, the reduced data rate required for the transmission of the audio component maximises the bandwidth available to the visual element.

The position today is that a number of low rate speech coding techniques have already been adopted as international standards for various network applications. For many new network developments such as global virtual private networks, third generation mobile, cellular satellite and even the asynchronous transfer mode (ATM) [10] networks, it is no longer a question of whether speech compression should be used, but which speech coding technology provides the target speech quality at lowest cost.

This paper highlights the major speech coding systems used in the existing networks, and summarises the activities in speech coding standards. The evolution of the key speech coding techniques is reviewed. Finally, a new class of speech coding technique, known as speech interpolation coding, is discussed. Recently, this type of speech coding technique has attracted much attention, and is believed to have the potential to provide good quality speech coding suitable for telephone use at or below 4 kbit/s.

## 2. Performance issues

The use of any speech processing system over the public switched telephone network (PSTN) is normally bounded by a set of stringent network planning rules in order to maintain a high quality of service and not to degrade existing applications. The principal aspects of performance against which a speech coder is assessed are [1, 11]:

- speech quality,
- delay,
- transparency to other voiceband signals,
- ability to support existing services using speech recognition or verification.

In addition, parameters such as speech quality and transparency are affected by a number of influences external to the coder, so it is important to take account of the following:

- dynamic range,

- robustness to channel errors,
- robustness to circuit noise,
- robustness to ambient background noise.

In general, the existing speech coding standards will have taken into consideration all of the above performance issues.

Table 1 lists some well-known international speech coding standards. The bit rate, algorithmic delay, subjective performance, and class of coding scheme are given for comparison. The bit rates indicated are the net speech coding rates excluding the error control coding used on the transmission channels. Algorithmic delay is purely the coding delay introduced by the fundamental speech encoding and decoding operations excluding the processing and transmission delay. The performance of a speech processing system is normally evaluated using mean opinion score (MOS), which is a formal subjective measure of received speech quality [12].

Generally, coding quality with MOS higher than 4 is considered as toll quality, between 3.5 and 4 as communication quality, between 3 and 3.5 as professional quality, and below 3 as synthetic quality [1, 13].

## 3. Speech coding in the network

### 3.1 National and international networks

The well-known 64-kbit/s PCM coding method of the ITU G.711 Recommendation, was standardised in 1972, but it was 1990 before the UK national backbone network finally became fully converted to this digital speech transmission technique. PCM coding provides toll-quality speech transmission, and is transparent to all the standard voiceband data transmission techniques.

As the national networks are normally owned and operated by the country's PTT organisation, low operating costs using simple PCM coding are more important than high bandwidth efficiency using more complex speech coding methods. In contrast, the international networks, such as networks for international direct dial (IDD) services, operate over undersea cables and satellites. These are costly shared resources, and therefore result in a more expensive service. A way to provide more efficient digital transmission in this situation is to combine the 64 kbit/s PCM coding with a technique known as digital speech interpolation (DSI) [14]. As normal conversational speech is active only about 40% of the time in either direction of transmission, a large group of simultaneous calls can be statistically multiplexed. The DSI suppresses the silent periods between the spoken words and syllables using

Table 1 Comparison of the well-known telephone-band speech-coding standards.

| Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Year   | Coding Type     | Bit rate (kbit/s)  | MOS*                  | Algor. delay (ms) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|--------------------|-----------------------|-------------------|
| ITU-G.711                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1972   | PCM             | 64                 | 4.3                   | 0.125             |
| ITU-G.721                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1984   | ADPCM           | 32                 | 4.0                   | 0.125             |
| ITU-G.726                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1991   | VBR-ADPCM       | 16, 24, 32 and 40  | 2.0, 3.2, 4.0 and 4.2 | 0.125             |
| ITU-G.727                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1991   | Embedded-ADPCM  | 16, 24, 32, and 40 | —                     | 0.125             |
| ITU-G.728                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1992   | LD-CELP         | 16                 | 4.0                   | 0.625             |
| Inmarsat-B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        | APC             | 9.6/16             | communication         | 20                |
| GSM Full-rate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1989   | LTP-RPE         | 13                 | 3.7                   | 20                |
| GSM-EFR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1995   | ACELP           | 13                 | 4.0                   | 20                |
| Skyphone                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1989   | BT-MPLPC        | 8.9                | 3.5                   | 28                |
| DAMPS Full-rate IS54                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1991   | VSELP           | 7.95               | 3.6                   | 20                |
| ITU-G.729                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1995   | CSA-CELP        | 8                  | 4.0                   | 15                |
| IS-96                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1991   | Qualcomm CELP   | 1, 2, 4, and 8     | 3.5                   | —                 |
| JDC Japanese Full-rate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        | VSELP           | 6.7                | communication         | 20                |
| GSM Half-rate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1994   | VSELP           | 5.6                | 3.5                   | 24.375            |
| ITU-G.723                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1995   | A/MP-MLQ CELP   | 5.27/6.3           | communication         | 37.5              |
| American DOD FS1016                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1990   | CELP            | 4.8                | 3.0                   | 45                |
| TETRA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1994   | ACELP           | 4.56               | communication         | 35                |
| ITU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (1998) | (To be defined) | 4                  | (toll)                | (25)              |
| Inmarsat-M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1990   | IMBE            | 4.15               | 3.4                   | 78.75             |
| JDC Japanese Half-rate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1993   | PSI-CELP        | 3.45               | communication         | 40                |
| American DOD FS1015                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1984   | LPC-10          | 2.4                | synthetic             | 22.5 (minimum)    |
| <p>* <b>WARNING:</b>The Mean Opinion Score (MOS) coding performance figures given above are obtained from different formal subjective tests using different test material, and they are evaluated under the most favourable listening conditions including the optimum speech input level, optimum listening level, no added background noise, no bit errors and no transcoding. The given MOS figures are therefore useful as a guide, but should not be taken as a definitive indication of codec performance. This measure is on a scale from 1 to 5, interpreted as follows: 1 = bad, 2 = poor, 3 = fair, 4 = good, and 5 = excellent.</p> |        |                 |                    |                       |                   |

reliable silence elimination and regeneration processes respectively at the transmitting and receiving terminals. While a talker is silent, the transmission channel is made available to the other active calls in progress. At the receive end of the link, 'comfort' noise is usually substituted during the periods of disconnection, so that the listener is less aware of the breaks in transmission. By sharing channels in this way, the capacity of the international links could be approximately doubled. However, if the system is not properly dimensioned, annoying speech-clipping effects may occur during busy periods due to speech bursts competing for the shared transmission resource. This is known as competitive clipping.

A more commonly used technique over international networks is to use DCME [8], which combines DSI and ADPCM speech coding to provide a compression gain of up to 5:1 for speech transmission. The ITU G.763 Recommendation is the international standard for DCME, and combines DSI with G.726 variable bit rate ADPCM.

Recommendation G.763 specifies that the DCME should always use 40-kbit/s ADPCM to transmit voiceband data, but speech is transmitted at 32 kbit/s, dropping to 24 or even 16 kbit/s when necessary. This variable bit rate feature for speech transmission aims to reduce the amount of competitive speech clipping under heavy traffic loading conditions, but, inevitably, the speech quality is degraded due to the reduced mean bit rate of the coded speech. It is possible to exercise control over the transmission performance of the system by setting a minimum mean speech coding bit rate objective. Typically, this would be about 3.6 bits/sample to ensure toll-quality transmission. If the mean bit rate falls below this value, new calls are prevented from gaining access to the system until the heavy loading subsides. This feature is known as dynamic load control (DLC). A block diagram illustrating the main DCME processes is shown in Fig 1.

An ITU 16-kbit/s speech coder, known as the G.728 low delay code excited linear predictive (LD-CELP) coder,

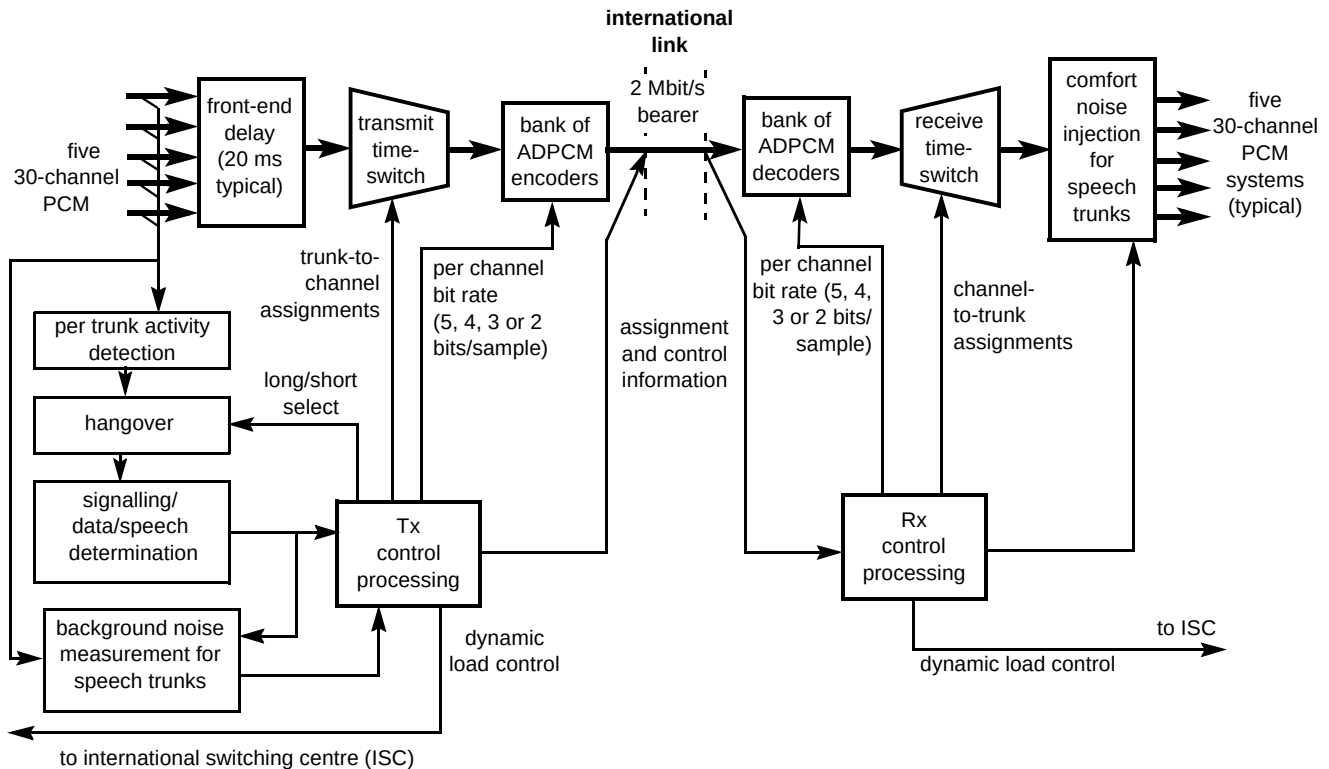


Fig 1 Block diagram to illustrate DCME processes.

became a standard in 1992, and demonstrated coded speech quality comparable to G.726 ADPCM at 32 kbit/s. A variable bit rate version of the LD-CELP coder is now being studied by the ITU. If this is used to replace the ADPCM coding in DCME, speech compression gain as high as 9:1 could theoretically be achieved.

Packetised networks, such as frame relay and ATM [10], are considered more flexible network technologies than the existing circuit-switched PSTN networks. A new ITU standard, the G.765 packet circuit multiplication equipment (PCME), has all the basic functions of a DCME, and more advanced features such as higher bit rate transmission (up to 150 Mbit/s) and the capability to support bandwidth-on-demand services. As this is a new standard, the use of PCME is not as popular as DCME, and it is viewed as a more advanced DCME for packet-switched networks. A well-known problem in packet transmission is packet loss during traffic congestion. To counteract this, PCME uses the ITU G.727 embedded ADPCM coding, which is similar to G.726 variable bit rate coding, but with the crucial difference that the variable bit rate aspect can be controlled within the transmission path without informing the encoder. With embedded coding, the transmitted values consist of 'core' bits essential for the signal to be decoded, plus 'enhancement' bits which contribute to the quality of the decoded signal. For each encoded sample, the enhancement bits may selectively be dropped to relieve congestion. In PCME, a block of speech samples is assigned to a packet, but, before transmission, the PCME rearranges the bits

within the packets according to their significance. When heavy traffic causes congestion, it is initially only the least significant part of a packet that is dropped, with progressively more significant parts being dropped as congestion increases. Thus the embedded coding property leads to a graceful decline in speech quality as traffic congestion rises [15].

Many manufacturers claim that DCME and PCME offer toll-quality transmission at high compression gain. However, the cost saving is route dependent and not directly proportional to the maximum compression gain of the equipment. The use of such equipment requires careful consideration of many factors, including cost of the equipment, cost of the international circuits, the traffic profile of the route, the relative proportions of speech and data traffic, system dimensioning, and the regulatory issues between the countries involved.

### 3.2 Mobile and satellite networks

It is already evident that digital public land mobile radio (PLMR) [16] systems are more robust, more flexible and offer a wider range of telecommunications services than analogue mobile systems. The development of digital PLMR has been the primary driving force in the past decade for the advances in speech coding technology that have been achieved. International mobile radio standards organisations, such as ETSI in Europe, and the Telecom-

munications Industry Association (TIA) in the United States, conduct comparative studies to choose the best speech coding systems available for their networks. Due to the demands for bandwidth efficiency and short timescales, and the limitations imposed by available technology, the standards organisations are usually prepared to accept a small compromise in the quality of the speech coding system in order to maximise the channel capacity. This near toll-quality speech is often referred to as communication quality.

The existing speech coding systems used in digital PLMRs are the 13-kbit/s long-term predictive, regular pulse excited (LTP-RPE) codec in GSM, the 7.95-kbit/s vector-sum excited linear predictive (VSELP) codec in the digital American mobile public system (DAMPS), and the 6.7-kbit/s VSELP codec in the Japanese digital communications (JDC) system. These codec bit rates apply to the basic, full-rate systems. Half-rate systems are achieved by halving the transmission bit rate of the radio link, and so doubling the channel capacity. Half-rate systems for GSM and JDC have just been completed, and the associated speech codecs are a 5.6-kbit/s VSELP codec for half-rate GSM and a 3.45-kbit/s pitch-synchronous innovation (PSI-CELP) codec for half-rate JDC. The subjective performance of these codecs (not the overall system performance) is presented in Table 1.

The above-mentioned mobile radio standards are not intended to provide services for maritime and aeronautical applications, nor even for remote or sparsely populated areas. Instead, Inmarsat has set up several standards and services to provide coverage in these areas via satellite. The well-known speech coding standards are the 9.6/16-kbit/s adaptive predictive coding (APC) for Inmarsat-B, the 9.6-kbit/s multipulse linear predictive coding (MPLPC) for Skyphone (otherwise known as Inmarsat-aero), and the 6.4-

kbit/s improved multiband excited (IMBE) coding for Inmarsat-M applications.

The second generation digital cordless telephone systems, such as CT-2 and the digital European cordless telephone (DECT) system, place the emphasis on low delay, good quality and inexpensive equipment rather than spectrum efficiency. ADPCM at 32 kbit/s, conforming to the G.726 standard, is the speech coding method adopted for these systems. A hybrid dual-mode system, DECT/GSM, to combine cordless and mobile applications into one handset is currently under study. The DECT system would be used in the indoor environment, automatically switching to the GSM system in an open area.

Applying speech coding over any radio network is generally much more difficult than on the fixed wireline network, since mobile radio users are usually in a noisy environment where the signal-to-background noise ratio could be as low as 10 dB. The speech codec must also face the harsh radio-fading effect which, in the GSM radio channel, can lead to a mean bit error rate as high as 10 per cent. The design of a speech coding system must be robust to these conditions.

Figure 2 shows the functional block diagram of a speech coding system typical of that used in digital mobile radio. The front-end of the speech encoding unit contains:

- a handset that provides good acoustic noise rejection,
- automatic gain control (AGC) to adjust the input level to the optimum operating range,
- analogue to digital conversion,
- an optional digital noise cancellation unit to further reduce the noise effect before encoding,

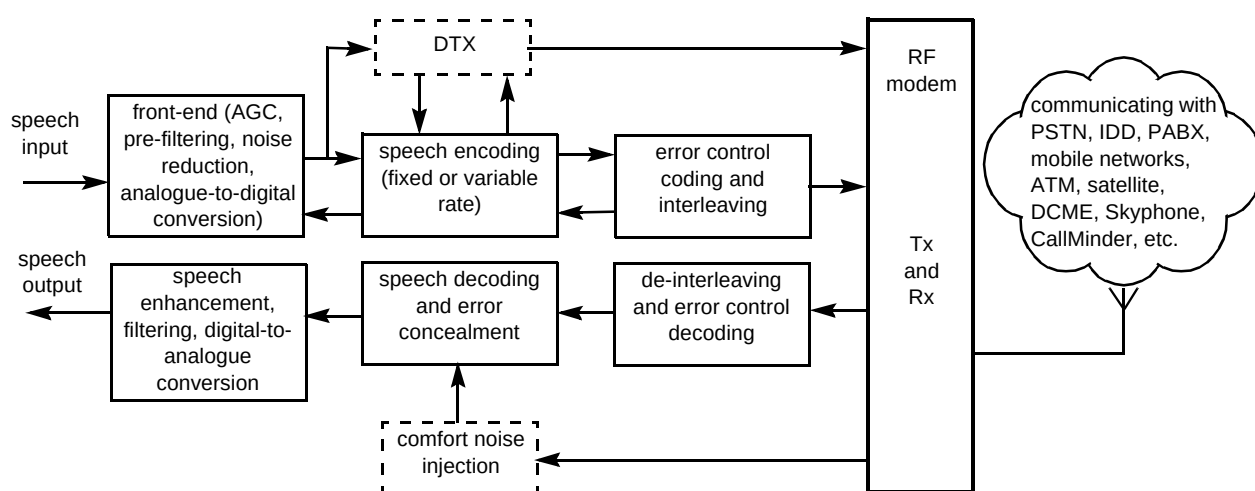


Fig 2 Block diagram of a typical mobile radio transceiver.

- pre-filtering or a signal modification process that aims to maximise the speech coding efficiency.

The coding performance could be further improved if all these functions are controlled by the speech encoding unit in a closed-loop manner based on some form of optimisation process [17].

After the pre-conditioning process, the speech signal is compressed by either a fixed or variable bit rate speech coding algorithm. Fixed bit rate coding is normally adopted if the speech coding and error control coding (channel coding) are designed separately. A more robust design approach, for a given gross channel bit rate, is to jointly optimise the speech and channel coding. The different component parts of the speech signal are encoded at their minimum acceptable rates, leading to a varying data rate (i.e. variable rate speech coding) [18]. The compressed speech is then error protected by different error-control methods which take into account the error sensitivity of the speech component and the remaining bits available for error protection [19, 20, 21]. The combined speech and channel coding can also be transmitted at variable rates; examples of this approach can be seen in the Qualcomm IS-96 speech coding standard.

Transmission errors will degrade the received speech quality, but the nature of the perceived distortion is dependent on the type of speech coder used. In general, as the speech coder bit rate reduces, the information content per bit becomes increasingly more significant. For the lowest bit rate speech coders, even the corruption of a single data bit could cause a severe speech degradation or result in loss of intelligibility. Fortunately, the error-control coding helps to maintain acceptable transmission performance under poor radio channel conditions by greatly reducing the number of digital errors presented to the speech decoder. Error-control coding, however, adds redundancy and interleaving delay to the protected data. Typically, the inclusion of error-control coding doubles the bandwidth relative to that required for the unprotected compressed speech, and the additional interleaving delay is significantly higher than the algorithmic delay of the speech codec.

At the receiver, the speech decoder performs the inverse of the encoding process to reconstruct the speech signal, but, error-control coding has its limitations. Under severe fading conditions residual errors will occur in the received data, causing transient distortions such as clicks and pops in the decoded speech, and badly degrading the quality of service. Error-concealment methods are used to suppress these transient distortions once channel errors are detected, and the corrupted speech signal is smoothly muted under long-term fading conditions. To improve robustness to residual errors, current research also attempts to use speech parameters with inherent error-detection capability [21] or

to optimise the bit representation for the coded speech parameters to reduce the effect of bit errors [22].

An optional function, known as discontinuous transmission (DTX) [6, 23], has been standardised for GSM to reduce co-channel interference and power consumption at the mobile terminal. This is achieved by transmitting only during the active periods of speech. The operation has similarities to DSI as described in section 2. The DTX feature has attracted considerable interest from other international mobile radio standards organisations. However, within GSM, the use of DTX is left to the discretion of the mobile radio network operator.

The basic functional blocks required for digital speech transmission over mobile and satellite channels have been described. The expected communication quality can be achieved and maintained only if these functions are properly designed and implemented [24].

Digital cellular systems are already in service, and as more emerge, it is becoming clear to the mobile users that the transmission performance is not comparable to the fixed wireline systems. Currently, ETSI has a new activity to develop a significantly better speech coder system for the existing full-rate GSM system. A new toll-quality 8-kbit/s coder, the G.729 conjugate structure algebraic (CSA)-CELP coder, was standardised by the ITU in late 1995. A primary application for this coder is to provide robust, toll-quality transmission for the future third-generation mobile systems.

### 3.3 Military

Very low bit rate speech coding at 2.4 kbit/s is in common use for military communications. Here, the objective is to provide robust transmission for various radio links even when subject to severe radio jamming. The coding scheme must also provide highly intelligible speech under unusual background noise conditions such as tank, helicopter, and machine-gun noise. As other aspects of speech quality are of less importance to military communications, even a synthetic quality coder might be acceptable if the objectives of high intelligibility and robustness to transmission conditions are achieved.

The American Department of Defense (DOD) plays an important role in the advancement of speech coding technology. Two well-known military standards are the LPC-10e 2.4-kbit/s vocoder based on linear predictive coding, and the FS1016 4.8-kbit/s CELP coder. The LPC-10e vocoder provides synthetic but intelligible speech quality at a very economical bit rate. The CELP coder became a standard in 1990, and the algorithm software is freely available on the Internet [25]. Several manufacturers have implemented the standard on various digital signal processing (DSP) devices. Although the speech quality is

not high, this codec has attracted much attention due to the potential for further improvements to the algorithm.

Currently, the DOD is the major driving force in developing a new 2.4-kbit/s speech coding standard, and recent results indicate that some 2.4-kbit/s coding methods already exceed the FS1016 4.8-kbit/s coding performance [26]. It is expected that even more significant improvement could be made by the time of the standardisation in mid 1996.

#### 4. Future demands

The previous sections have highlighted the necessity for low rate speech coding in present-day communications networks. In the future, speech coding can be expected to play a vital and expanding role in many areas.

For mobile networks, state-of-the-art speech coding will continue to be crucial, but the objectives will become even more demanding as these networks grow and customer expectations increase. In the near future, the objective for the enhanced full-rate GSM system is to provide transmission performance at least as good as the fixed wireline network. This will also be the objective for third-generation mobile systems [24].

If a good quality mobile service could be provided at sufficiently low cost, the number of domestic mobile customers would be expected to increase dramatically. The present norm of one fixed telephone per household might well become one mobile phone for each member of the family! This will not be achieved without extremely efficient use of radio bandwidth, leading to a requirement for speech coders at even lower bit rates than those used today. The ITU has an objective to standardise a toll-quality 4-kbit/s speech coding system in 1998 [11]. The target application areas for this codec are mobile, satellite, and very low rate visual telephony. For a number of satellite applications such as Inmarsat-P [27], and Iridium [28], there are also requirements to develop and use 2.4-kbit/s speech coding.

The future will also see an expansion of telephone voice services outside the traditional person-to-person conversation call, and low rate speech coding will play a crucial role in the economic provision of many of these services. The recent introduction of 'CallMinder' [5] into the national network is an example of the type of interactive voice messaging service destined to become commonplace. CallMinder is a sophisticated call-answering and message-storage facility, network-based so that no additional equipment is required at the customer's home or office. Its advanced capabilities allow two incoming calls to be handled at once, even if the customer is already on the phone, and for each customer, up to thirty separate messages may be stored. The role that low bit rate speech

coding plays in this service is clear, since to store numerous voice messages using 64-kbit/s PCM would make the system prohibitively expensive; 1 Mbyte of storage would afford only about two minutes of recorded speech.

CallMinder uses a low bit rate speech coder to reduce the storage costs. The time needed to retrieve the speech data is also reduced, and the data transmission requirements within the equipment are eased.

Archiving conversational speech is important for legal purposes in many spheres of work, such as business meetings, telephone banking, betting shops, police-questioning, insurance companies, and the legal profession. Speech compression will be increasingly exploited for this purpose. With high-quality 2.4-kbit/s speech coding operating with a VAD, about 2.5 hours of coded speech can be stored on a low-cost 1.4-Mbyte high-density floppy disk. The storage costs are significantly cheaper than for analogue tape recording. As digital speech also allows the use of additional control software, selected speech passages can be very quickly retrieved from the storage system for listening, or for transfer to a document by an audio-typist. Sometimes, a very large speech database is required to be transferred between different systems; with a compressed speech database, communications costs are cheaper and the transfer time is shorter.

High-quality low bit rate speech coding could also be used to compress the speech database used for real time text-to-speech systems [29] and speech recognition, leading to more cost-effective solutions.

Multimedia is another area destined to become a part of everyday life. The combined use of low bit rate speech and visual coding will mean that multimedia services can be made available over the PSTN. The PC videophone (VC8000) hardware developed by BT is an example of this technology, and permits multimedia communications over ISDN networks.

An emerging packet speech transmission technology allows full-duplex conversation over the Internet. There are already a number of products — usually referred to as Internet phone or Iphone [30] — on the market, and they all use low bit rate speech coding to maximise their transmission performance over the Internet medium. Packet voice transmission over the Internet [31] will suffer the same transmission problems as those described in section 3.1 for PCME. However, the Internet is not engineered to cope with the requirements of voice traffic. The transmission delay will be variable, and unacceptably high when the Internet link is heavily loaded. For current Internet voice products, the quality of service cannot be guaranteed. It will be interesting to see whether these systems ever become more than the equivalent of a 'CB radio' channel for the Internet amateur.

## 5. The technology

Over the past three decades, many different types of speech coder have been developed. For PSTN applications, speech quality and delay are the key performance issues, but these must be balanced against the cost of implementation. The simplest form of coding in the PSTN is the toll-quality G.711 64-kbit/s PCM coding standardised in 1972. Today, an 8-kbit/s CSA-CELP coder has been shown to provide near toll-quality coding, and is due to become the ITU G.729 standard in 1995 — speech compression by a factor of eight achieved in little more than two decades. Although the G.729 codec is much more complex than the G.711 codec, the advances in microelectronics allow it to be implemented at low cost.

There are four main classes of speech coder, these being waveform coding, vocoding, hybrid coding, and frequency domain coding. These classifications are fully described in Boyd [1]. Within each class there are many types and variants, each having their own attributes. This section does not attempt to describe every known speech coding method, but concentrates on highlighting the key techniques used in speech coding standards.

### 5.1 Memoryless coding

This is the simplest form of coding, and G.711 64-kbit/s PCM coding is an example of this method. Each input signal sample is rounded (or quantised) to the nearest representative level, independent of the past or future nature of the signal. The quantised levels may either have a uniform separation, or a non-uniform (typically pseudo-logarithmic) arrangement as in the case of G.711 PCM. For a uniform quantiser, the coding distortion is directly proportional to the variance (or the energy) of the input signal. If the input signal to the uniform quantiser is adjusted to the full operating range, the signal-to-quantisation noise ratio, SNR, is estimated as  $6R$ , where  $R$  is the number of coding bits per sample. However, for a non-uniform quantiser, the lower signal levels are encoded with finer precision, leading to a more consistent coding performance over the dynamic range. The SNR of this non-uniform quantiser can be approximated as [32]:

$$SNR_{PCM} = 6R - 10 \text{ (dB)}$$

For the 8-bit non-uniform quantiser of G.711, this corresponds to an SNR of about 38 dB for either A-law or  $\mu$ -law PCM coding, and its subjective performance is considered to be good, achieving a MOS rating normally above 4.

### 5.2 Adaptive differential predictive coding

The distortion produced by a quantiser is proportional to the signal variance, and its subjective performance degrades rapidly as the bit rate is reduced. It has been observed that speech is a quasi-stationary signal in the short term, i.e. the

difference amplitude between adjacent speech samples has much lower variance than the sample amplitude itself. This characteristic of a speech waveform is exploited in differential PCM (DPCM) and its derivatives such as ADPCM.

The DPCM coder uses a predictor to estimate the amplitude of the next input speech sample from the history of past samples. It is the difference between this estimate and the actual sample amplitude which is coded. With an appropriate predictor, this difference signal will be small and therefore have lower variance than the actual signal. The better the predictor, the smaller the variance. This variance-reduced signal, usually known as the prediction error or residual signal, can then be quantised at a lower PCM coding rate. The amount of the variance or energy reduction is known as prediction gain,  $G_p$ . The overall SNR of a predictive PCM coding system [32] is given by:

$$SNR = SNR_{PCM} + G_p \text{ (dB)}$$

Two approaches are possible to the design of an adaptive predictor. The prediction can either be derived using the history of the quantised signal, which is known as backward prediction, or using a segment of the uncoded input speech, which is known as forward prediction. A higher prediction gain can normally be achieved by the forward prediction method, but the input signal must be buffered and the predictor parameters are transmitted as side information. The buffering delay introduced for forward prediction is usually the dominant part of the algorithmic delay introduced by the coder.

In ADPCM, which is the speech coding technique used in the ITU G.726 and G.727 standards, the predictor is made adaptive to the continuously changing characteristics of the speech waveform, leading to a more accurate prediction and, hence, increased prediction gain. To satisfy the low delay requirement, the G.726 and G.727 [33] coders use a 2-pole and 6-zero adaptive backward prediction method to encode the signal on a sample-by-sample basis. These coders also use adaptive quantisers which attempt to match the quantiser range to the short-term power of the signal being coded. As the signal level reduces, the quantiser range contracts, and vice versa. Because of this, the separation between the discrete quantiser levels is reduced for low-level signals, allowing these signals to be encoded with finer precision. At 32 kbit/s, the SNR for the decoded speech signal is about 22 dB which is about 16 dB lower than for 64-kbit/s PCM coding, but the subjective MOS performance is normally about 4, only slightly lower than for 64 kbit/s PCM. This highlights the difficulty of using objective measures to quantify the performance of a speech coder.

In recent speech coder designs, the performance of the adaptive prediction has been further improved by exploiting both the short-term and the long-term nature of the speech signal [34]. Many techniques have been proposed to derive



these predictors, but linear predictive analysis is the most commonly used in speech coding. The short-term predictor normally aims to model the vocal tract function, and the long-term predictor to model the pitch periodicity of the speech signal. For 4-kHz bandwidth speech, a 10th order (or 10 taps) short-term predictor and a 1-tap long-term predictor are usually used and, typically, they together provide a prediction gain of between 8 and 15 dB. As this combination of predictors has proved to be robust and practical to implement, they have been used in many digital mobile radio coding standards.

However, the achievable prediction gain depends on the types of predictor used and the nature of the speech signal to be coded. The prediction gain does not increase linearly as the number of taps is increased, and tends to level off at high orders of prediction.

### 5.3 Analysis-by-synthesis vector quantisation

Instead of coding the speech sample or the prediction error on a sample-by-sample basis such as in the G.711 PCM and the G.726 ADPCM coders, higher coding efficiency can be achieved if a set of related parameters, or a vector of elements, is encoded to a representative vector extracted from a pre-stored codebook of optimised vectors. This is known as vector quantisation (VQ) [35]. Even better coding performance can be achieved if VQ is combined with an analysis-by-synthesis (AbS) method [36, 37]. The AbS VQ method forms the basis for coding the prediction error in code-excited linear predictive (CELP) coding. The basic difference between the direct VQ and the AbS VQ method is in the definition of the quantisation distortion measure used in the VQ codebook search.

For example, the 16-kbit/s ITU G.728 LD-CELP coder uses a 50th order backward linear prediction method, and a 10-bit vector codebook to encode a vector of 5-sample prediction error. The direct VQ approach would select the stored vector that gives minimum quantisation distortion on the 5-sample prediction error, while the AbS VQ method used in G.728, and illustrated in Fig 3, searches for the vector that gives minimum quantisation distortion between the input and the coded speech. As the coded speech is computed from the prediction error through a filtering process and this has to be performed on every codebook vector to be tested, the AbS VQ method is much more complex than the direct VQ method.

The general drawbacks of VQ are the large memory needed to store an optimised vector codebook at the encoder and at the decoder, and the high computational complexity to determine the best vector via an exhaustive codebook search. For these reasons, systems using VQ coding are more expensive to implement than those using PCM or ADPCM coding methods.

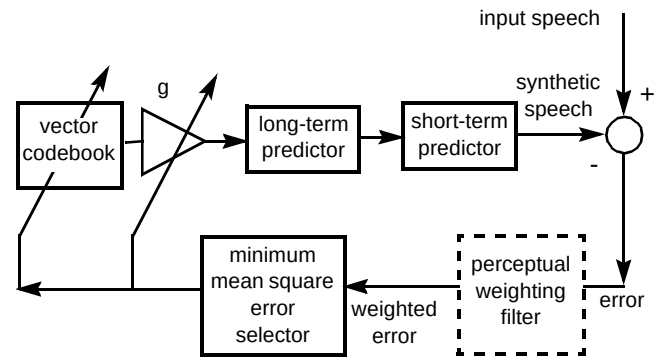


Fig 3 AbS-VQ in CELP coding.

For a speech signal, the measured SNR of the 16-kbit/s LD-CELP coder is about 18 dB, which is about 20 dB lower than for the G.711 64-kbit/s PCM coder. The ITU 8-kbit/s CSA-CELP coder uses forward prediction and the AbS VQ method to encode the prediction error. The measured SNR is about 13 dB.

### 5.4 Noise shaping and masking

Inevitably, even with the advanced speech coding technology described above, the signal-to-quantisation noise ratio decreases as the coding rate decreases. It has been observed [38] that the perceived quantisation noise is affected by the spectral energy of the speech. A speech signal usually exhibits a large variation in the frequency domain, as illustrated in Fig 4, but the frequency spectrum of quantisation noise is normally evenly spread across the frequency band. This frequency varying nature of the SNR suggests that the perceived noise in the high spectral energy regions, such as the speech formant regions, is not as significant as in the low spectral energy regions, such as the trough regions.

Based on this principle, a technique known as perceptual weighting or noise shaping, aims to reduce the perceived noise by shaping the noise in the frequency domain so as to improve the SNR in the trough region at the expense of reducing the SNR in the formant region. In fact, a perceptual weighting filter (PWF) is used in all CELP coding standards. The PWF is incorporated with the AbS VQ in the encoding process, as shown in Fig 3, so that the quantised prediction error vector selected from the codebook would minimise the perceptually weighted error between the input and the coded speech. An example of the frequency response of a PWF is shown in Fig 4.

At the decoder, the perceived quantisation noise may be further reduced by a post-filtering process [39], also known as 'speech enhancement' as in the case of the Inmarsat-M IMBE codec. The technique exploits the noise-masking effect in human perception. One signal is said to be masked when its audibility is reduced by the presence of another, louder signal at the same or near frequency. An example of

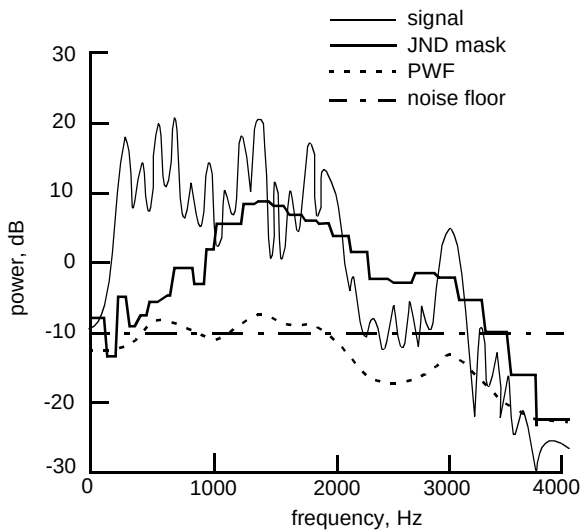


Fig 4 Noise shaping for speech coding.

masking is the failure to hear a telephone ringing when listening to loud music. The basic principle of post-filtering used in CELP coding is to artificially strengthen the formants and the peaks of the pitch harmonics so that quantisation noise between the peaks is partially or totally masked. This technique becomes increasingly important at the lower speech coding rates.

However, since post-filtering modifies the coded signal, consideration must be given to the possibility of codecs being operated in tandem. If a post-filter is not carefully designed or disabled in the tandeming situation, the coded signal may be modified several times leading to degraded performance.

### 5.5 Latest developments in low rate coding

Combining the latest coding technology and good engineering, the 8-kbit/s G.729 CELP coder has achieved near toll-quality comparable to the G.726 32-kbit/s ADPCM coder. Based on current evidence, it seems unlikely that CELP coding could achieve near toll-quality coding at 4 kbit/s or below to satisfy the future demand mentioned in section 4. The primary limitation is the performance when coding voiced speech such as the vowel sounds. Three major themes in speech coding research work are currently being actively pursued, namely, efficient VQ, auditory modelling, and coding methods for voiced speech.

The use of VQ is computationally expensive and tends to be signal dependent, but it is a crucial technique for very low bit rate coding. The future research objectives are to develop low-complexity VQ techniques which are robust to different types of background noise, frequency/attenuation distortion of the input signal, and channel errors.

Noise shaping and masking, as discussed in section 5.4, have been successfully used in speech coding. The trend is to exploit more general masking properties of the human hearing process to derive a dynamic, frequency-dependent threshold, below which noise will be perceptually inaudible. This is sometimes referred to as the just noticeable distortion (JND) threshold, and it is already widely used in wideband audio coding [40]. An example of this threshold and its corresponding signal spectrum are given in Fig 4, and clearly, it has a very different characteristic to the more 'speech-specific' PWF threshold. In principle, as long as the coding noise is maintained entirely below the JND threshold — even with noise higher than the signal in some spectral regions — transparent coded speech performance could be achieved. The use of JND or its combination with perceptual weighting and post-filtering would make a significant contribution to achieving toll-quality coding at very low bit rates.

Toll-quality coding of voiced speech has been a difficult objective for low bit rate speech coding. Recent developments show that more effective coding of these sounds can be achieved via interpolation of representative waveforms. As voiced speech is a fairly stationary signal, it consists of near replicas of itself when observed in a short time window. When coding speech below 4 kbit/s, a 'representative' waveform segment can be extracted regularly and efficiently coded; the signal between the coded segments can then be regenerated via interpolation. With an appropriate speech interpolation process, this method can result in lower overall quantisation noise than competing techniques. Several interpolation methods have been proposed in recent years. This paper classifies this type of technique as speech interpolation coding, and the more popular approaches are summarised in the next section.

## 6. Speech interpolation coding

The use of time-domain interpolation is a common feature of many of the speech coding techniques now being proposed for very low bit rates. The principle is to characterise speech by analysing short representative segments extracted at suitable time intervals. The characteristics of each analysis segment are assumed to be 'instantaneous' measurements which describe the speech waveform in the vicinity of a particular point in time referred to as an update point. These characteristics are efficiently represented by encoded parameters. At the decoder, frames of speech are synthesised with smoothly changing parameters obtained by interpolating between the encoded parameters. Synthesis frames begin and end at update points and, in general, the lengths of analysis segments and synthesis frames will be different. The interpolation process is intended to approximate the changes that occur in the original speech, from one update point to the next.

Among the techniques actively studied in recent years for very low bit rate speech coding are the following which use time-domain interpolation:

- sinusoidal transform coding (STC) [41, 42],
- multiband excitation coding (MBE) [43, 44],
- prototype waveform interpolation (PWI) [45].

### 6.1 Analysis and coding

Each of the three techniques mentioned above may be implemented with update points at regular intervals of about 20 ms. At each update point, an estimate of the pitch-frequency (the fundamental frequency of the vocal cord vibration) is required for the speech waveform in the vicinity of the update point when it may be considered voiced. The pitch frequency is considered variable, and the estimate obtained is taken as an 'instantaneous frequency' at the update point. Some quite sophisticated pitch-frequency estimation techniques are required as this is a critical aspect of all three techniques.

**Sinusoidal transform coding (STC)** is based on a method of representing speech as the sum of time-varying sinusoids whose amplitudes, frequencies and phases vary with time and are specified at regular update points. The method was first applied to low bit rate speech coding [41] in 1985. At the analysis stage, segments of the original speech, each of duration about two and a half pitch-periods and centred on an update point, are spectrally analysed via a fast Fourier transform (FFT). For voiced segments, the power spectra will have peaks, in principle, at harmonics of the pitch-frequency as illustrated in Fig 5. Unvoiced segments will have randomly distributed peaks. The frequencies of the peaks are identified by applying a simple peak-picking algorithm and the corresponding magnitudes and phases are then obtained. These measurements become the parameters of the STC representation. It has been found [42] that speech synthesised from these parameters can be made essentially indistinguishable from the original speech when the parameters are unquantised.

For low bit rate coding, it is not possible to encode, with sufficient accuracy, the STC model parameters directly. An indirect method is adopted whereby a spectral envelope represented by a small number of parameters, a 'sign (ambiguity) bit', a pitch-frequency measurement and a 'voicing probability' frequency are encoded at each update point. The sign bit is needed, as  $s(t)$  and  $-s(t)$  have the same spectral envelope. The voicing probability frequency divides the speech spectrum into two bands; a lower band considered voiced and an upper band considered unvoiced. This frequency, and the sign-bit are determined by an 'analysis-by-synthesis' procedure to determine its voicing decisions. A means of deriving the sinusoidal model parameters from this information at the decoder will be discussed in the next section.

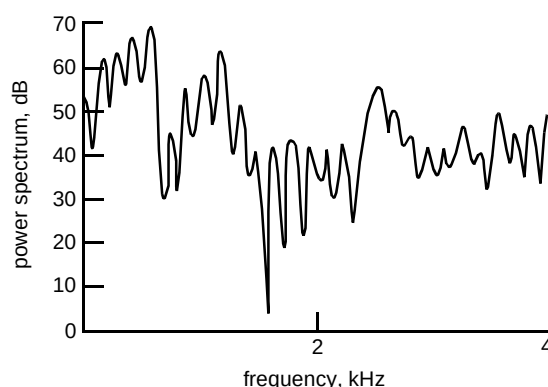


Fig 5 Power spectrum modelled by STC.

**Multiband excitation (MBE)** coding was originally proposed [43] for 9.6-kbit/s coding and later improved and adapted to 4.15 kbit/s, referred to as the IMBE codec. The IMBE codec, combined with error protection to yield an overall 6.4 kbit/s, is used for Inmarsat-M satellite mobile communications [44]. MBE coding involves extracting parameters at the update points from relatively large analysis segments of fixed length which overlap as illustrated in Fig 6.

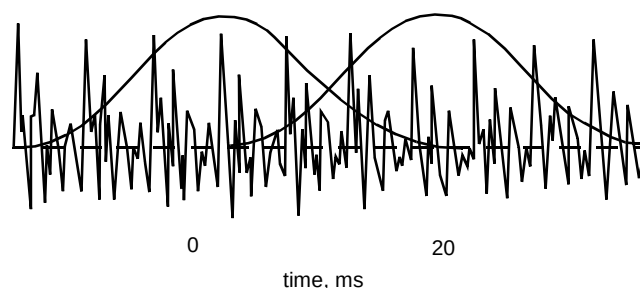


Fig 6 Overlapped windows used by IMBE.

A two-stage pitch-period extraction procedure is applied, the first stage achieving an accuracy of one half of a sampling interval. The second stage refines the result of the first stage to an accuracy of one quarter of a sampling interval. From the first pitch-period estimate, non-overlapping frequency bands are defined, each of bandwidth equal to a specified number of pitch-frequency harmonics — typically three. The number of bands is therefore pitch-period dependent and may lie between 3 and 12. A separate voiced/unvoiced decision is next made for each of these bands by measuring the similarity between the original speech in the band and the closest approximation that can be synthesised by harmonics of the more accurately estimated pitch-frequency. When the similarity is close, the band is declared voiced — otherwise it is classified as unvoiced. Three 'spectral amplitudes' are determined, corresponding either to the pitch-frequency harmonics

within the band or, for an unvoiced band, to samples of the speech spectral envelope. The input speech is now assumed to be characterised at each update point by a pitch-period estimate from which the pitch frequency and the number of frequency bands may be determined, and for each of these bands, a voiced/unvoiced decision and three spectral amplitudes. This information is then efficiently coded for transmission.

**Prototype waveform interpolation (PWI)** was proposed [45] in 1991 for speech coding at 3 to 4 kbit/s. A prototype waveform is a segment of voiced speech of length equal to one pitch period, starting at any point within a pitch cycle. The original idea of PWI was to encode separately the lengths and shapes of such waveforms extracted, as illustrated in Fig 7, at update points which lie within voiced

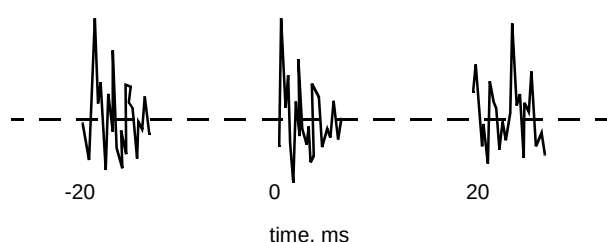


Fig 7 Prototype waveforms for PWI

portions of the speech. It was proposed that a voiced/unvoiced decision should be made for each update point, and that unvoiced segments should be encoded by switching to a form of CELP. Each prototype waveform is characterised by passing it through an LPC inverse filter to generate a spectrally flattened residual. The residual is then analysed to produce a Fourier series:

$$e(t) = \sum_{k=0}^{P/2} \left[ A_k \cos \left( \frac{2\pi kt}{P} \right) + B_k \sin \left( \frac{2\pi kt}{P} \right) \right]$$

whose period  $P$  is equal to the estimated pitch-period. Over the analysis interval,  $e(t)$  is equal to the prototype waveform residual. 'Time alignment' is applied to  $e(t)$  to make the  $A_k$  and  $B_k$  coefficients as close as possible to the  $A_k$  and  $B_k$  coefficients of the previous update point. The resulting  $A_k$ ,  $B_k$  and linear predictive coefficients (LPC), the pitch period  $P$ , and a one-bit voicing decision are quantised and efficiently encoded for voiced speech.

Quantisation distorts not only the shape of each prototype waveform, but also the way this shape changes from one update point to the next, i.e. the waveform 'dynamics'. The result may be too much, too little and/or inappropriate periodicity in the decoded speech. An important innovation of PWI is the use of a 'signal-to-change ratio' (SCR) in the quantisation procedure. This ratio measures the rate at which the periodicity is changing in the original speech, and attempts are made to quantise the parameters in such a way that this rate of change is

preserved at the decoder, even when the decoded waveforms are considerably different from the original.

Recently [46], the concept of a prototype waveform has been generalised to include arbitrary length segments of unvoiced speech. The term 'characteristic waveform' is now used. Instead of extracting a single characteristic waveform at each update point, a sequence of about ten are extracted at regular intervals between consecutive update points. When these waveforms are time aligned, as illustrated in Fig 8, the changes that occur to their corresponding Fourier series coefficients are indicative of the nature of the speech. Rapid changes occur for unvoiced speech, slow changes for voiced speech. High- and low-pass digital filtering is applied to separate the effect of these changes, and the resulting filtered Fourier series coefficients characterise a 'slowly evolving waveform' (SEW) and a 'rapidly evolving waveform' (REW) which sum to form the true characteristic waveform coefficients. The SEW is down-sampled to one waveform per update point, rapid fluctuations having been eliminated by the low-pass filtering. The REW cannot be accurately represented at low bit rate. Fortunately, it may be replaced at the decoder by a random waveform with similar spectral shape. The parameters for this generalisation of PWI, i.e. LPC coefficients, pitch period and characteristics of the SEW and REW, may be encoded at 2.4 kbit/s.

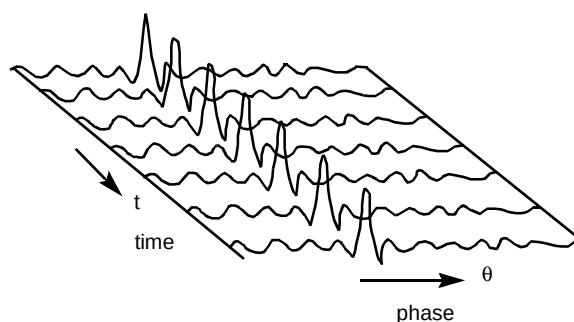


Fig 8 Evolution of prototype waveform.

**Comparison of analysis and coding techniques** — as described above, all three techniques may be considered as sinusoidal coding techniques, where fundamental differences lie in the lengths of the analysis segments. For MBE, the relatively large analysis segment gives good spectral resolution and noise immunity. However, the effects of non-stationarities, such as that due to pitch-frequency variation, will be apparent in the resulting spectra. These effects are accommodated to a considerable extent by the MBE coding technique. A PWI analysis segment, being of length equal to a single pitch period, may be expected to give a more accurate representation of voiced speech at an update point and be less affected by pitch-frequency variation and other non-stationarities. The variable length of the analysis segment adopted by STC is not a critical factor and is chosen to allow an accurate

spectral envelope to be determined without serious distortion caused by non-stationarity. Vector quantisation is used liberally by all three techniques, e.g. for LPC coefficients, spectral amplitudes, SEW and REW. The principles of all three techniques are still the subject of active research.

## 6.2 *Decoding and synthesis*

All three techniques synthesise speech at the decoder as sums of sinusoids whose amplitudes, instantaneous frequencies and phases are changing on a sample-by-sample basis and are obtained by interpolating between values obtained from decoded parameters at each update point.

A straightforward linear interpolation scheme is used for the amplitudes, which must change monotonically across the synthesis frame.

The term 'instantaneous frequency' is used for a frequency modulated sine wave to describe its apparent frequency at a given instant in time. Voiced speech conforms more naturally to this sinusoidal model than unvoiced, and for all three techniques, the instantaneous frequencies at each update point are simply the harmonics of the decoded pitch frequency. There are important differences in the way each of the three techniques adapt the model to unvoiced frames.

**STC** — the STC decoder receives a pitch frequency measurement, a set of parameters representing the spectral envelope, a 'voicing probability' frequency and a sign bit. Below the voicing probability frequency, the amplitudes of the sinusoidal model are obtained at each update point by sampling the decoded envelope at the pitch-frequency harmonics. The phase of each sinusoid is not available directly from the decoded parameters, but is deduced from the spectral envelope on the assumption that it is the gain response of a minimum phase transfer function. STC models speech above the voicing probability frequency by sinusoids closely spaced in frequency (say 100 Hz apart) synthesised with random phase. Again the amplitudes of these sinusoids are obtained by sampling the decoded envelope. The sign bit applied to each sinusoid will eliminate the possibility of random 180° phase reversals from one update point to the next.

As the phases of the sinusoids are specified or estimated at the update points, the instantaneous phase of each sinusoid is computed using cubic interpolation for a more faithful reconstruction of the original speech waveform.

**IMBE** — although earlier versions of MBE encoded phase information, the improved version (IMBE) encodes only spectral amplitude measurements for each band. These determine the amplitudes of (normally) three sinusoids for each voiced band, and the power and spectral shape of the required band-limited random signal generated for each

unvoiced band. The number of bands and the frequencies of the sinusoids are deduced from the decoded pitch period.

IMBE makes no attempt to model the true phase relationships between sinusoids at the update points; instead the phases at the update points are simply chosen to maintain continuity with the waveform synthesised in the previous frame. This is achieved by linear interpolation from frame to frame of the corresponding pitch frequency harmonics. It can be shown that linear interpolation of a harmonic frequency is equivalent to a quadratic interpolation of its phase, but this form of interpolation introduces arbitrary phase shifts. IMBE only applies this method for relatively stationary voiced harmonics. It is known that disregarding the true phase relationships in this way entails some loss of naturalness, and the original waveshape is not necessarily preserved.

A less complex but a reasonable approximation to the effects of interpolation is obtained by an 'overlap and add' technique, used for harmonics with large variation in pitch frequency and for transitions in either direction between voiced and unvoiced conditions. Centred on each update point, frames are initially reconstructed without interpolation using sinusoids and/or random signals derived from the coded parameters. Between two update points, the current and previous reconstructed frames are merged to form the final synthesised speech frame. To merge the frames, the current frame is multiplied by a gradually increasing window, and is then added to the previous frame multiplied by a gradually decreasing window.

**PWI** — the original version of PWI is restricted to voiced portions of speech, unvoiced portions requiring a switch to CELP. As the sine and cosine amplitudes of the Fourier series are encoded, the correct phase relationships between pitch frequency harmonics may be preserved. However, pitch synchronism with the original speech is not maintained because the prototype waveforms are time aligned by the encoder for efficient quantisation. Further time alignment is needed at the decoder to maintain the continuity of the synthesised waveform. This is achieved by equating the phases at the update point to the instantaneous phases attained at the end of the previous synthesis frame.

PWI employs linear interpolation of the pitch period, which is broadly equivalent to quadratic interpolation of the phase information except that the use of sine as well as cosine terms in the Fourier series preserves the original phase relationships between harmonics. A synthesised waveform similar to the original, but with an imperceptible time shift, which varies from frame to frame, is characteristic of PWI encoded waveforms.

## 6.3 *The way forward*

Three basic forms of speech interpolation coding have been briefly summarised. These coding methods have enabled great advances to be made in the field of low bit

rate coding for telephone bandwidth speech, and are still an active research topic. Although the development of the new 2.4-kbit/s speech coder for the American DOD standard has shown very promising results with these coding methods, it is not yet certain whether these methods could achieve toll-quality coding for the future ITU 4-kbit/s coding standards. A number of questions need to be resolved.

Do the simple rule-based interpolation methods (linear, quadratic or cubic) impose a barrier to toll-quality coding? Could these methods adequately track the speech dynamic behaviour even within a short speech segment (typically about 20 ms)? Would these artificial rule-based interpolation methods tend to introduce more synthetic decoded speech under codec tandeming conditions?

Although some speech signals look fairly stationary, some frequency regions could have larger variations at certain 'time zones'. It may be desirable to adaptively allocate encoded bits to adjust the interpolation strategy for the frequency components at the time of large variations. Could better coding efficiency be gained with such an adaptive time-frequency bit allocation scheme?

Some 'representative' speech waveforms may be formed as a combination of two or more complex waveforms; they may be subject to different time evolution from the previous representative waveform. Does toll-quality coding require careful control of different interpolation functions over these complex waveforms?

At present it seems possible that there is much potential for refinement and cross-fertilisation of the principles behind the three coding techniques reported here. New and practical ideas are being reported all the time. However there is also the possibility of a totally different approach revolutionising this field of research yet again.

## 7. Network interactions

This paper has highlighted the general performance issues, the bit rates, system complexities and key techniques of speech coding systems used, or destined to be used, in the telephone network. Low bit rate speech coding has undoubtedly proved beneficial in many network applications, but it also imposes limitations. The fundamental reasons for these limitations, and the techniques used to evaluate and resolve the consequent network issues, are discussed below.

The telephone's history was dominated by essentially linear systems [47]. Amplitude quantisation, as in PCM coding, is commonly believed to be the only nonlinear process used in speech coding. For low bit rate coding as discussed in section 6, quantisation can, in fact, come in various different forms or in combinations of these. The STC coder quantises the frequency spectrum into a set of spectral peaks — frequency domain quantisation. The STC and PWI coders quantise a long speech segment as a representative speech waveform — time domain

quantisation. Noise masking coders quantise speech into perceptually important signal components — 'hearing' quantisation. Time domain interpolation also introduces 'artificial' time variation into the decoded signal. These different types of time variant nonlinear processes are important for achieving high-quality speech coding at low bit rates, but this means that white granular quantisation noise is no longer the only speech coding distortion that exists in the network.

As a result of these nonlinearities, when low bit rate coders are used, problems may arise due to adverse interactions [24, 48, 49] with other elements of the telephone network. For example, echo canceller performance may be impaired, and high bit rate voiceband data transmission may become unworkable over the link. If codecs of the same or different type are connected in tandem, unacceptable distortion may result. It is therefore necessary to give consideration to possibly problematic configurations, so that appropriate advice can be given to network planners.

BT Laboratories have facilities to evaluate an extensive range of network configurations by both objective and subjective test methods. A portable, modular DSP platform developed at the Laboratories and known as the network emulator [48] is routinely used to facilitate these investigations. Software emulations exist for all of the common network elements, and selected emulations are downloaded to the DSP hardware to provide a real time emulation of the chosen network configuration. When it is desired to evaluate the performance of a particular codec in a network scenario, this is usually achieved by converting the codec algorithm into a software module for the network emulator. It may then be tested in any network configuration. Modules exist for many of the codec standards, e.g. G.711, G.726, G.728, GSM, and Skyphone. A recent addition to the library of software modules is that of DCME to CCITT Recommendation G.763. This module allows any DCME trunk-to-channel ratio and loading to be emulated, and will continually output important system statistics such as mean bit rate, and the amount of speech clipping. Since DCME involves the combination of voice activity detection, DSI, comfort-noise injection, and variable bit rate ADPCM, there is considerable scope for adverse interactions to occur when other low rate codecs form part of the link. The DCME module will allow such connections to be evaluated, and field problems to be investigated.

As the bit rate decreases, the design of speech coders will become more 'speech specific', and they will be less likely to offer transparent transmission for voiceband data such as modem and facsimile data, or even in-band network signalling tones, such as dual-tone multifrequency (DTMF). These disadvantages can be resolved using reliable signal classification methods to identify speech, modem data,

facsimile and network tones. The different signal types can then be transmitted using different speech coding rates or techniques such as modem or facsimile demodulation [50].

## 8. Conclusions

Contrary to popular belief, G.711 64-kbit/s PCM coding is no longer the only digital transmission method in the switched telephone network. It still reigns supreme in the PSTN trunk network due to the full-scale deployment of optical fibre, and the relatively cheap bandwidth this affords. However, low rate speech coding is in common use in international, mobile, satellite and private networks, and these form a major part of the overall network. In digital mobile radio systems, low rate coding offers greatly improved channel capacity — indeed, these systems would not be viable without it. In fixed networks including satellite, international and private networks, low rate coding is in high demand because of the substantial cost savings. Low rate coding is also essential to allow multimedia services over conventional telephone lines. In this context, the digital superhighway will become less necessary as compression technology improves.

Utilising the latest speech coding technology, the new ITU G.729 8-kbit/s coder achieves toll-quality speech coding performance. The application of this codec could significantly improve the transmission performance of existing mobile radio networks such as full-rate GSM.

Recent research results show that speech interpolation combined with noise masking speech coding techniques for achieving toll-quality performance at 4 kbit/s. The ITU already has a plan in place to achieve this goal by 1998.

As coding performance improves at lower bit rates, speech codecs will become widespread in the switched network. With a variety of speech coders used in different network scenarios, the likelihood of tandemed codecs will increase, and the need to guard against adverse network interactions will become increasingly relevant. If a connection comprises more than one speech codec, the transmission delay and distortion will increase significantly so that echo control may be required. This is a particular case where careful planning is vital to prevent serious degradation occurring. For the network operators, a good understanding of network interactions with low rate coding systems will become an important issue both to guarantee a quality service to customers and to ensure that none of the existing services are degraded.

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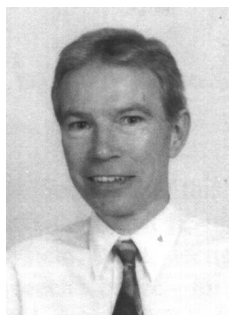
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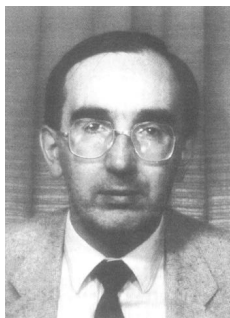
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